

CITS3402/CITS5507 Assignment 1: Parallel Hash Collision Attack

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Birthday-attack algorithm and collision-detection data structure

`toy_hash` produces a 48-bit output, so by the birthday bound a matching pair between two independent nonce-trial families is expected after roughly $\sqrt{\pi/2 * 2^{48}} \sim 2.1e7$ combined trials — far fewer than 2^{48} for a preimage search. Rather than fixing one file’s nonce and searching only the other, the program grows two trial pools in parallel, stopping once a trial’s hash already exists in the *other* file’s pool.

Both pools live in one shared open-addressing hash table keyed by the 48-bit hash. Each slot stores the hash, the nonce that produced it, which file it came from, and a “ready” flag. Insertion mixes the hash with a Fibonacci multiplier, masks it to the table size to pick a home slot, then linear probes to the first free or matching slot. A match owned by the *other* file is a genuine collision; one owned by the *same* file is a redundant self-collision, discarded. Capacity is four times the trial budget ($4 * 2^{26}$ slots) so load factor stays at or below 0.5 and probe chains stay short. Nonces are generated sequentially via an atomic counter per file rather than randomly — `toy_hash`’s MurmurHash3-style finalising mix already scatters sequential inputs uniformly, so this behaves like random sampling while wasting no nonce on a repeat.

Parallelisation and synchronisation with OpenMP

Threads split evenly into an “A-group” and a “B-group” (even/odd thread id) in one `#pragma omp parallel` region; each hashes trials only for its file, drawing the next nonce with `atomic_fetch_add` on a per-file counter — no two threads ever hash the same nonce, and none blocks.

The shared hash table is the only structure threads communicate through, made safe without any `critical` section or `omp_lock_t`. Each slot’s hash field is a C11 `_Atomic uint64_t` written only via `atomic_compare_exchange_strong` from `TABLE_EMPTY` to the trial’s hash; exactly one thread can win that CAS per slot, so that slot’s `nonce/owner` fields are subsequently written by that thread alone — no two threads ever write the same memory. The remaining hazard is a reader observing a slot as claimed before the winner finishes writing those fields; this is closed with a “published” flag — the winner writes `nonce/owner`, then stores `ready = 1` with `memory_order_release`, and a reader spins on `ready` with `memory_order_acquire` first, so it never observes a torn write. That spin window is only the few instructions between the CAS succeeding and the stores after it, so it costs nothing in practice. A second atomic (`found_flag`) is claimed with one final CAS so only the first thread to find a collision records the winning pair; every other thread observes the shared `stop_flag` and exits. Before writing any file, the program recomputes `toy_hash` on the final bytes and aborts on disagreement, so a concurrency bug can never silently produce an invalid submission.

Memory requirements and trade-offs

The dominant cost is the collision table: at the default 2^{26} -trials-per-side budget, capacity is 2^{28} slots, each holding an 8-byte hash, 8-byte nonce, and two 1-byte tags — about 4.3 GB. This beats a smaller, higher-load-factor table since linear probing degrades sharply near load factor 1, and Kaya has far more memory than this needs, so speed wins over economy. A smaller cost is per-thread: each thread keeps one private, mutable copy of whichever file(s) it hashes (loaded once, header patched every trial), so only the 16-byte nonce is rewritten per trial rather than re-copying the whole file — for the largest (~900 KB) pair at 96 threads this is under 90 MB, negligible beside the table.

Performance metrics and analysis

`scripts/scaling_job.slurm` measured `search_seconds` on Kaya (`cits3402` partition, 96-core node) across all six pairs and thread counts 1–96, three repeats each, with a fixed per-pair trial budget (~8 s at 1 thread) isolating raw throughput from the attack’s own luck. Scaling is close to linear even for the hardest pair:

threads	search_seconds (6_exa)	speedup
1	36.10	1.00x
2	18.05	2.00x
4	9.02	4.00x
8	4.51	8.00x
16	2.26	15.98x
32	1.13	31.84x
64	0.58	62.70x
96	0.40	91.37x

That is 95% parallel efficiency at 96 threads. Every other pair scales just as well — 88.3% (`3_giga`) to 95.2% (`6_exa`) — and slightly *improves* with file size, since a larger per-hash cost leaves threads relatively less time touching the shared table, the one serialisation point. This far exceeds a 10-core laptop’s ~63% efficiency ceiling at 10 threads: Kaya’s server-class memory subsystem sustains many more concurrent accesses into the table before threads contend for bandwidth.

Full end-to-end solves on Kaya (96 threads, ID 24914408, all verified against `check_toy_hash.py`) confirm every pair completes well inside the 900 s budget:

pair	search_seconds	trials/side	% of budget
1_kilo	34.18	25.30M	3.8%
2_mega	30.66	11.41M	3.4%
3_giga	62.82	13.36M	7.0%
4_tera	74.45	10.58M	8.3%
5_peta	179.19	17.84M	19.9%
6_exa	693.55	36.83M	77.1%

Trial count is a random variable around the ~2.1e7-combined birthday-bound mean, not a function of file size — `2_mega` needed under half the trials of `1_kilo` despite twice the per-hash cost — so

wall-clock time must be measured per pair. `6_exa` is tightest at 77% of budget: its per-hash cost is roughly 14x `1_kilo`'s for a comparable trial count, exactly why harder pairs need the full 96-core node rather than fewer threads.